KENYA CERTIFICATE OF BASIC EDUCATION (K.C.B.E)

GRADE 10 ESSENTIAL MATHEMATICS TERM 1 FEBRUARY 2026

PATHWAY: S.T.E.M, SOCIAL SCIENCE, ARTS AND SPORT

Time: 2 Hours

COMPETENCY BASED EDUCATION CKEAB 002

LEARNER'S DETAILS

Name: _____.

School: _____.

Assessment Number: _____. **Date:** _____.

School Code: _____. **Signature**: _____.

INSTRUCTIONS TO LEARNERS

1. *Answer ALL questions in this paper*
2. *Show all working clearly. Marks will be awarded for method.*
3. *Non-programmable calculators may be used.*
4. *All answers should be written in the spaces provided.*
5. *Take π = 3.14 or 22/7 where necessary.*
6. *Study all diagrams, graphs, and images carefully before answering.*

Level	Level	% Range	Points	Learner's score
Exceeding (EE)	EE1	90–100%	8	
	EE2	75–89%	7	
Meeting (ME)	ME1	58–74%	6	
	ME2	41–57%	5	
Approaching (AE)	AE1	31–40%	4	
	AE2	21–30%	3	
Below (BE)	BE1	11–20%	2	
	BE2	1–10%	1	

SCORE GRID

ROW / QN	1 a	1 b	1 c	2	3	4	5	6	7	8	9	10	11 a	11 b	12	13	14	15	16	17 a	17b (i)	17b(ii)	17b(iii)	17 c	18 a	18 b	19 a	19 b	19 c	19 d	20a(i)	20a(ii)	20 b	20 c	21 a	21 b
Marks	2	3	1	4	3	3	3	3	3	3	3	1	2	3	3	3	3	3	3	2	1	2	2	5	5	1	4	2	3	1	2	4	3	5	5
Learner's Score																																				
Performance Level																																				

DESCRIPTION	SCORE
TOTAL SCORE	100
LEARNER'S SCORE	
PERFOMANCE LEVEL	

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K.C.B.E CBC CBE SBA SENIOR SCHO

ESSENTIAL MATHEMATICS (100 Marks)

Answer all questions in this section.

1. *A farmer divides his land among his three children as shown in the information chart below.*

Information chart:

i. *First child receives 2/5 of the land*

ii. *Second child receives 1/3 of the land*

iii. *Third child receives the remainder*

If the Total land area is 450 hectares

a) *Calculate the fraction of land received by the third child. Show your working.* ***(2 marks)***

b) *Calculate how many hectares each child receives.* ***(3 marks)***

c) *Express the ratio of land distribution among the three children in simplest form.* ***(1 mark)***

2. Use logarithm tables in all your steps to evaluate: leaving your answer to four decimal places (4mks)

$$\sqrt[3]{\frac{38.32 \times 12.964}{86.37 \times 6.285}}$$

3. When a certain number **x** is divided by 30, 45 or 54, there is always a remainder of 21.

Find the least value of the number **x** (3mks)

4. Find the value of **x** (3mks)

$2^{(x-3)} \times 8^{(x+2)} = 128$

5. Evaluate without using a calculator (3mks)

$$\frac{\left(1\frac{3}{7}-\frac{5}{8}\right)\times\frac{2}{3}}{\frac{3}{4}+1\frac{5}{7}\div\frac{4}{7}\,of\,2\frac{1}{3}}$$

6. Without using tables and calculators, evaluate

$$\sqrt[3]{\frac{0.032+0.0608}{1.28\times0.4}}$$ (3mks)

7. Use tables of squares, square roots and reciprocals to evaluate correct to 4 s.f

$$\frac{3}{\sqrt{0.0136}} - \frac{2}{(3.72)^2}$$ (3mks)

8. Simplify the following expressions by reducing it to a single fraction

$$\frac{2x-5}{4} - \frac{1-x}{3} - \frac{x-4}{2} \quad \text{(3mks)}$$

9. Sixteen men working at the rate of 9hrs a day can complete a piece of work in 14 days.

How many more men working at the rate of 7 hours a day would complete the same job in 12 days?(3mks)

10. Solve the equation;(3mks)

$$9^{x+1} + 3^{2x+1} = 36$$

11, A triangle ABC is formed by the points A (3,4), B (-7,2), and C (1,-2).

(a) Find the coordinates of the mid-points k of AB and p of AC (1 mk)

(b) Find the equation of the perpendicular bisector of the line kp (2 mks)

12. Two similar cans have different heights 8cm and the other one 10cm. If the surface area of the larger can is $480cm^2$, find the surface area of the smaller can. (3 marks)

13.Find the value of x that satisfies the equation log (2x – 11) – log 2 =log 3 – log x(3 marks)

14. A metal hemisphere of radius 12cm is melted done and recast into the shape of a cone of base radius 6cm. Find the perpendicular height of the cone (3mks)

15. Solve the inequality and show the solution on the number line. (3mks)

$3-2x \angle x \angle 2x+5$

16 .The angle of elevation of the top of a tree from a point P on the horizontal ground is24.5°. From another point Q, five metres nearer to the base of the tree, the angle of elevation of the top of the tree is 33.2°. Calculate to one decimal place, the height of the tree (3mks)

17.Three town P,Q and R are such that P is on a bearing of 120° and 20 km from Q. Town R is on bearing of 220° and 12km from P

a) Using a scale of 1 cm to 2 km, draw and locate the positions of the three towns. (3mks)

b) Measure

i) the distance between Q and R in kilometres. (2mks)

ii)the bearing of P from R. (1mk)

iii)the bearing of R from Q. (2mks)

c)Calculate the area of the figure bounded by PQR. (2mks)

18.The information below shows the entries in a surveyor's field book after a survey of a farm.

XY = 280m is the baseline. All measurements are in metres

	280	**Y**
B 105	230	110E
	190	
	160	45E
A 100	90	
	40	95G
X	O	

(a) Use a scale of 1cm represents 20m to draw the map of the farm (5mks)

(b) Estimate the area of the farm in hectares (5mks)

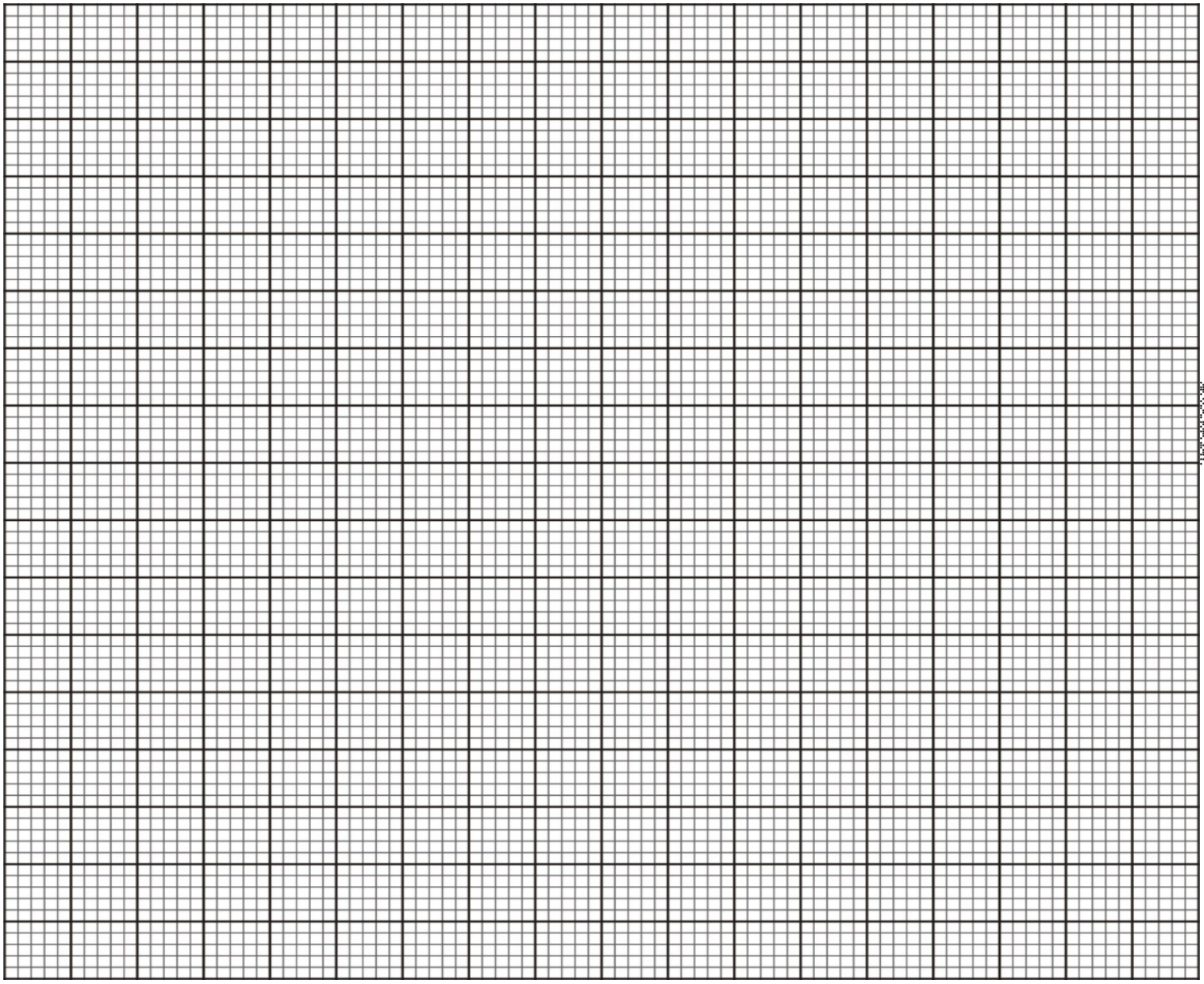
19. Triangle PQR has vertices P(3,2), Q(-1,1) and R(-3,-1).

(a) Draw PQR on the grid provided. (1mk)

(b) Under a rotation the vertices of $P^1Q^1R^1$ are $P^1(1,4)$, $Q^1(2,0)$ and $R^1(4,-1)$. Find the centre and angle of rotation using points P and Q. (4mks)

(c) Triangle PQR is enlarged with scale factor 3 centreO(0,0) to give triangle $P^2Q^2R^2$. Draw triangle $P^2Q^2R^2$ and state its co-ordinates. (2mks)

(d) Triangle $P^1Q^1R^1$ undergoes reflection in line y = -x to give triangle $P^3Q^3R^3$. draw $P^3Q^3R^3$ and state its coordinates.



20 The results of a mathematics test that a hundred students took are as shown below:-

Marks	No. of students
30-34	4
35-39	6
40-44	10
45-49	14
50-54	X
55-59	24
60-64	14
65-69	6

(a) Determine

(i) the value of X (1mk)

(ii) The modal class(2mks)

(b) Calculate the mean(4mks)

(c) The median (3mks)

21.A country bus left Nairobi at 10.45a.m and traveled towards Mombasa at an average speed of 60km/h. A matatu left Nairobi at 1:15p.m on the same day and traveled along the same road at an average speed of 100km/h. The distance between Nairobi and Mombasa is 500km.

(a) Determine the time of the day when the matatu overtook the bus (5mks)

(b) Both vehicles continue towards Mombasa at their original speeds. How long had the Matatu waited before the bus arrived? (5mks)

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